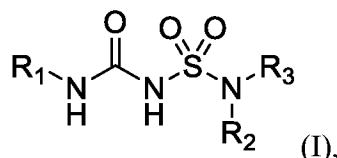


CLAIMS:

1. A compound of Formula (I):



or a solvate, or pharmaceutically acceptable salt thereof, wherein:

R₁ is C₃-C₁₆ cycloalkyl optionally substituted by one or more **R_{1S}**, wherein the C₃-C₁₆ cycloalkyl is a C₃-C₈ monocyclic cycloalkyl or polycyclic cycloalkyl, or, **R₁** is a phenyl substituted by two or three **R_{1S}**; wherein each **R_{1S}** is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₁-C₆ alkoxy, C₁-C₆ haloalkoxy, or halo, and at least one **R_{1S}** is C₁-C₆ alkyl, C₁-C₆ haloalkyl, or halo;

R₂ is $-(\text{CX}_2\text{X}_2)_n\text{-R}_{2S}$, wherein n is 0, 1, or 2, and each **X₂** is independently H, C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl, wherein the C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl is optionally substituted with one or more halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo;

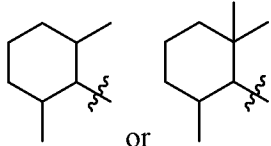
R_{2S} is 4- to 8-membered monocyclic heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo; and

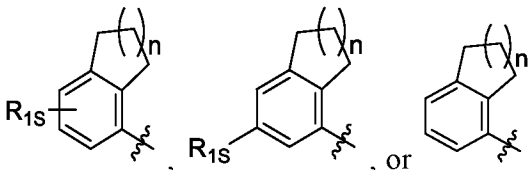
R₃ is 7- to 12-membered heterocycloalkyl or 5- or 6-membered heteroaryl optionally substituted with one or more **R_{3S}**, wherein each **R_{3S}** is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN.

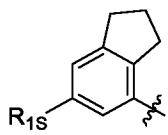
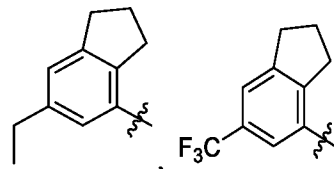
2. The compound of claim 1, wherein **R₁** is C₃-C₇ monocyclic cycloalkyl, C₉-C₁₀ bicyclic cycloalkyl, C₁₂-C₁₆ tricyclic cycloalkyl, or phenyl, wherein the C₃-C₇ monocyclic cycloalkyl, C₉-C₁₀ bicyclic cycloalkyl, or C₁₂-C₁₆ tricyclic cycloalkyl is optionally substituted by one or more **R_{1S}**, the phenyl is substituted by two or three **R_{1S}**; and wherein at least one **R_{1S}** is C₁-C₆ alkyl, C₁-C₆ haloalkyl, or halo;

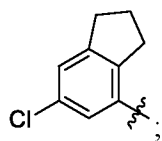
optionally wherein:

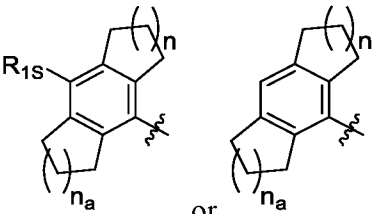
(a) the C₃-C₇ monocyclic cycloalkyl is cyclopentyl, cyclohexyl, or cycloheptyl, wherein the cyclopentyl, cyclohexyl, or cycloheptyl is optionally substituted by one or more **R**_{1s}; further

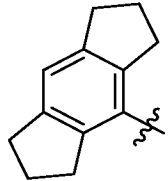
optionally wherein the C₃-C₇ monocyclic cycloalkyl is  or ;

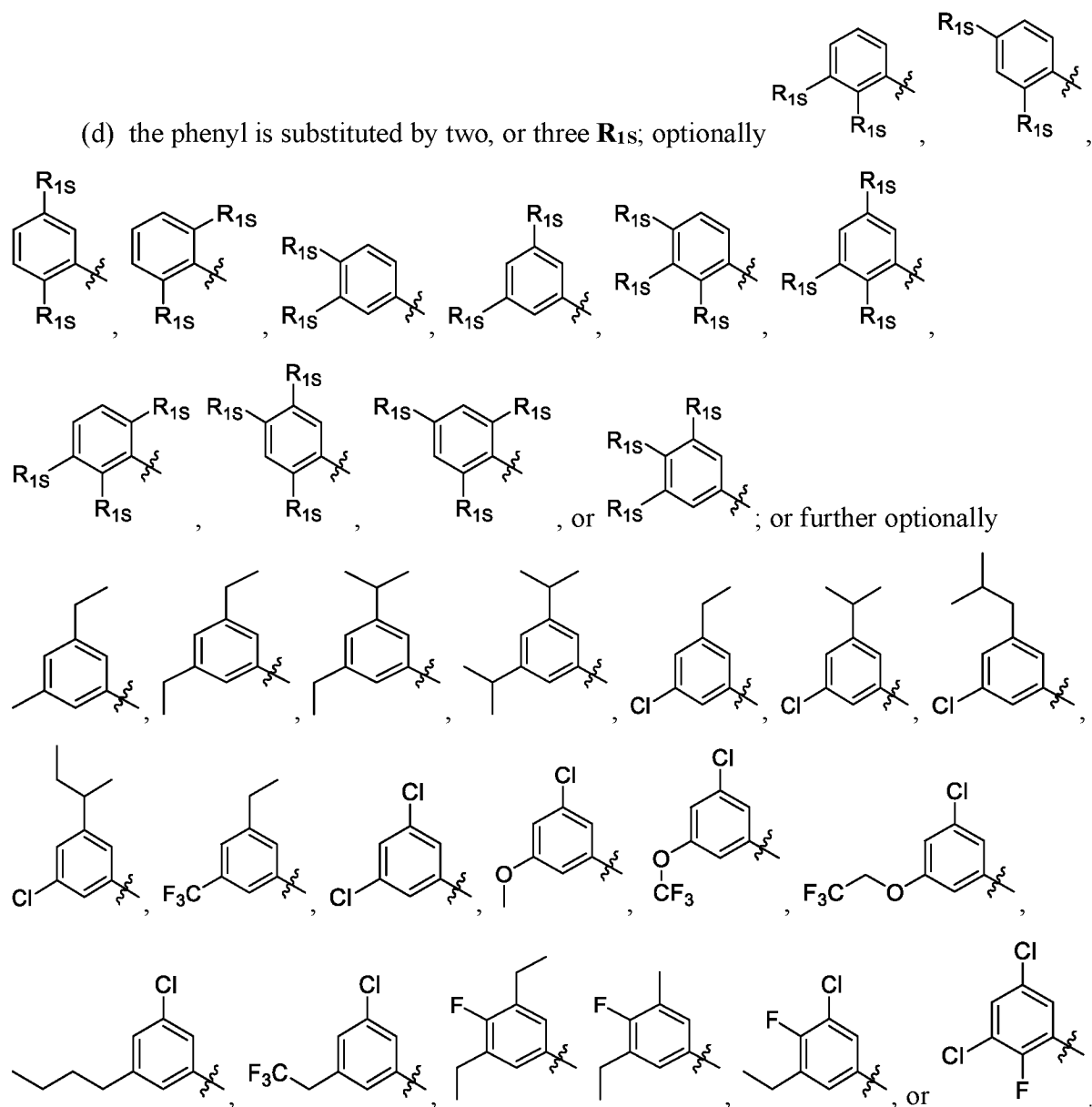
(b) the C₉-C₁₀ bicyclic cycloalkyl is , wherein n

is 0, 1, or 2; optionally ; further optionally , or



(c) the C₁₂-C₁₆ tricyclic cycloalkyl is , wherein n and n_a

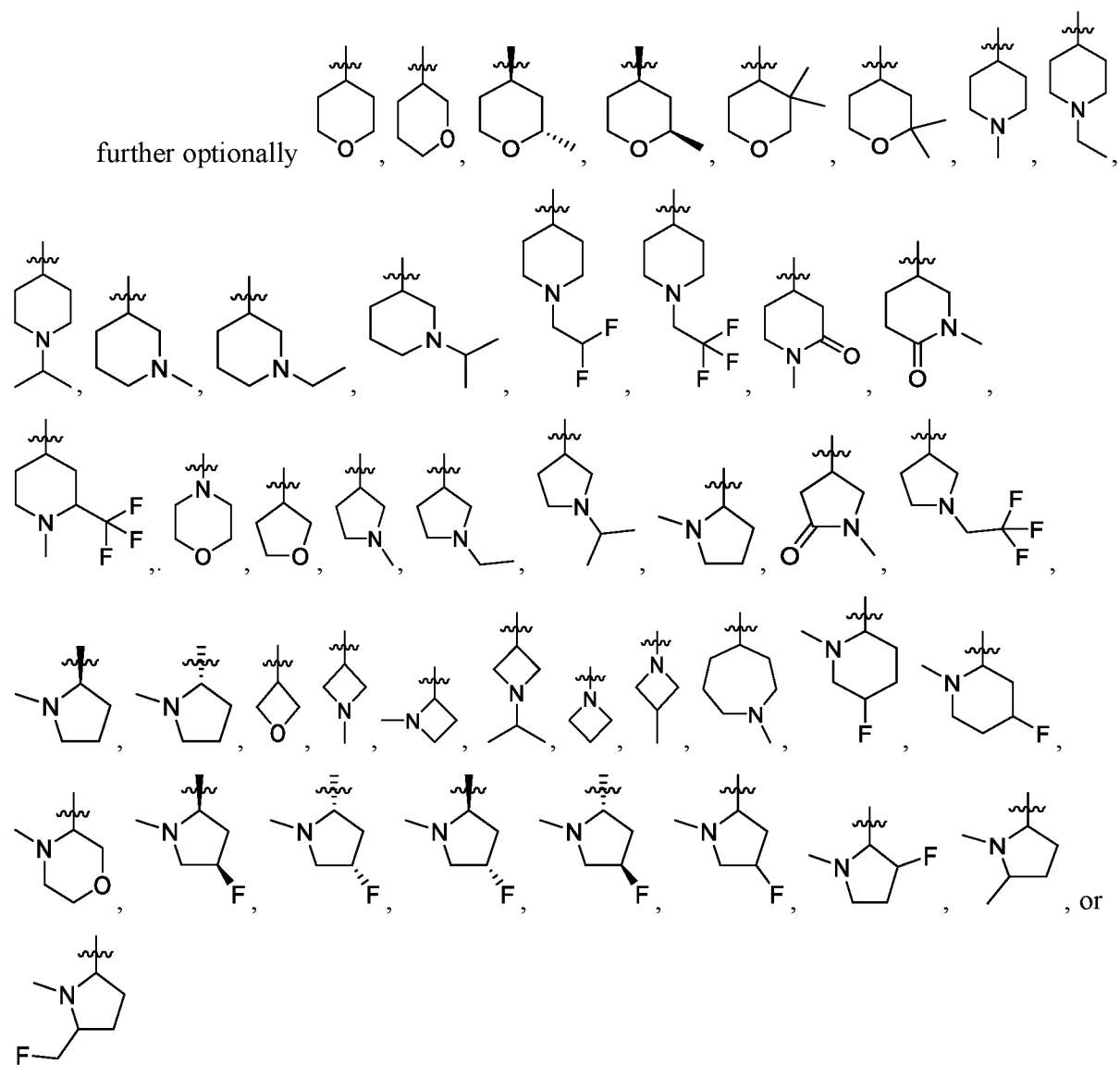
each independently are 0, 1, 2, or 3; further optionally ; and/or



3. The compound of claim 1 or 2, wherein R_2 is $-R_{2s}$, $-(CX_2X_2)-R_{2s}$, or $-(CX_2X_2)_2-R_{2s}$.

4. The compound of any one of claims 1 to 3, wherein R_{2s} is 4- to 8-membered monocyclic heterocycloalkyl optionally substituted with one or more C_1 - C_6 alkyl, C_2 - C_6 alkenyl, C_2 - C_6 alkynyl, C_1 - C_6 haloalkyl, halo, $-CN$, $-OH$, $-O(C_1$ - C_6 alkyl), $-NH_2$, $-NH(C_1$ - C_6 alkyl), $-N(C_1$ - C_6 alkyl) $_2$, or oxo;

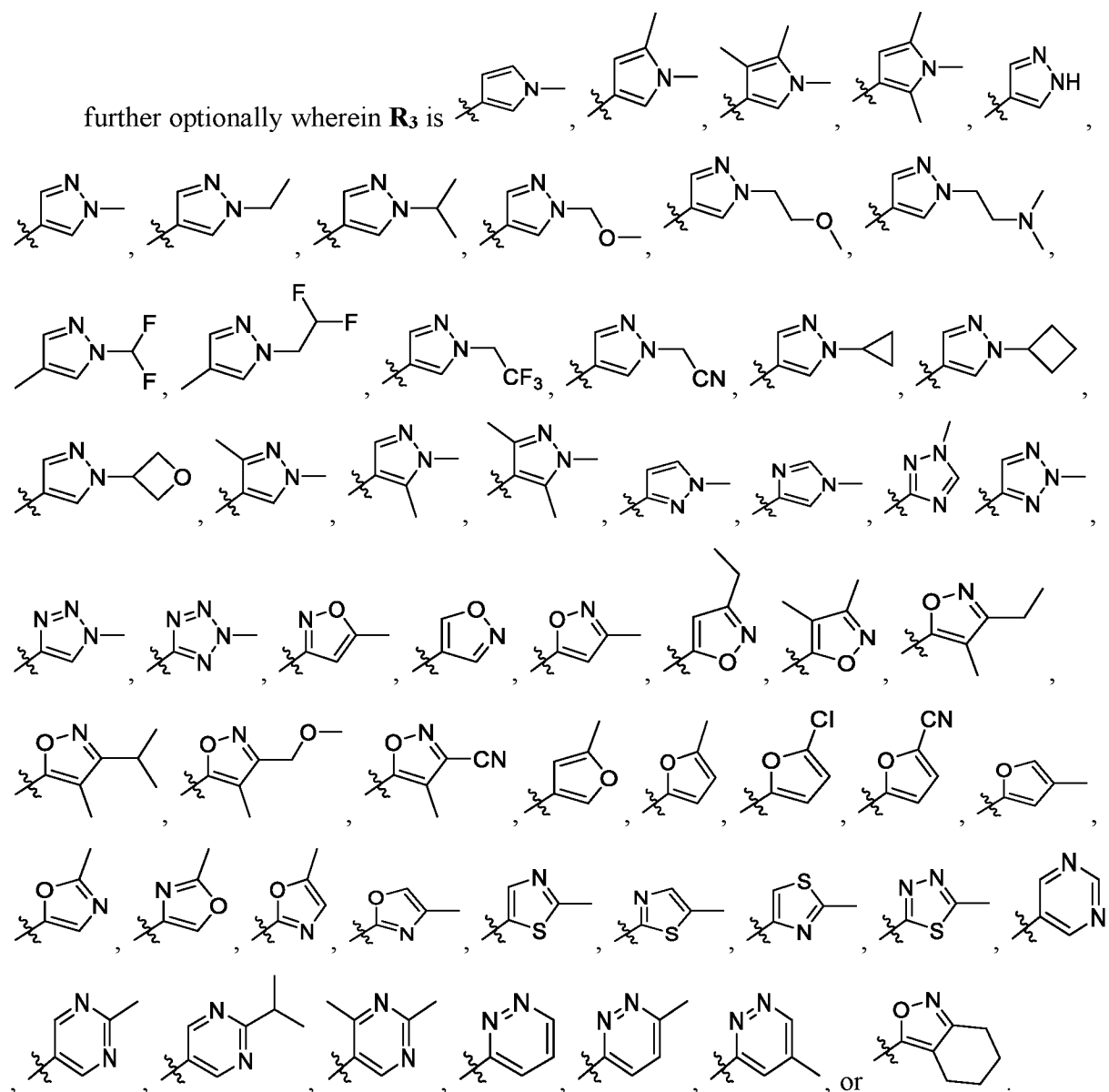
Optionally oxetanyl, azetidiny, tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, morpholinyl, or azepanyl, wherein the oxetanyl, azetidiny, tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, morpholinyl, or azepanyl is optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo; or



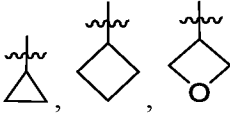
5. The compound of any one of claims 1 to 4, wherein **R**₃ is 7- to 12-membered heterocycloalkyl optionally substituted with one or more **R**_{3S};

optionally wherein **R**₃ is pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-

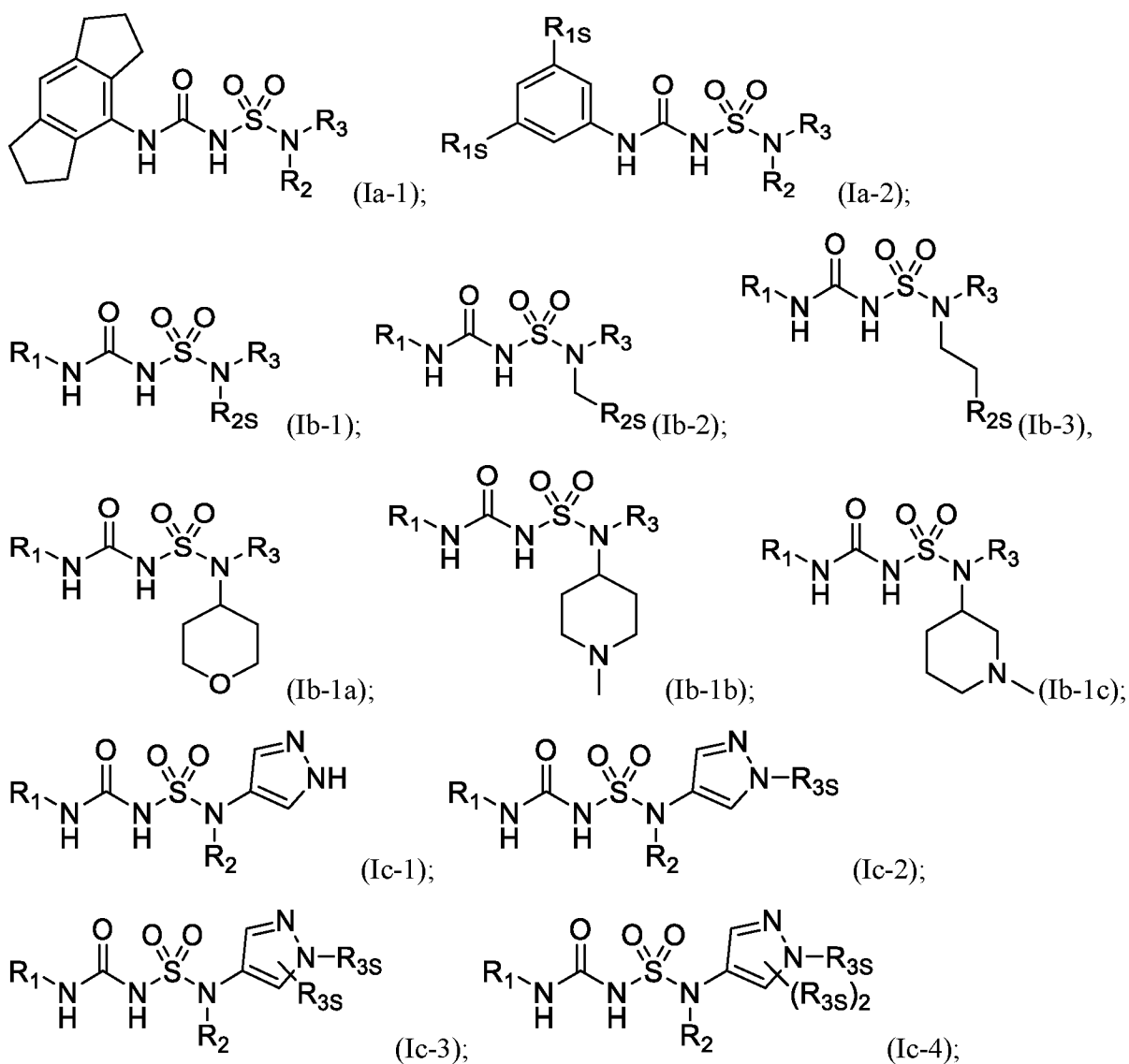
tetrahydrobenzo[c]isoxazole, wherein the pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-tetrahydrobenzo[c]isoxazole is optionally substituted with one or more **R**_{3S}; or

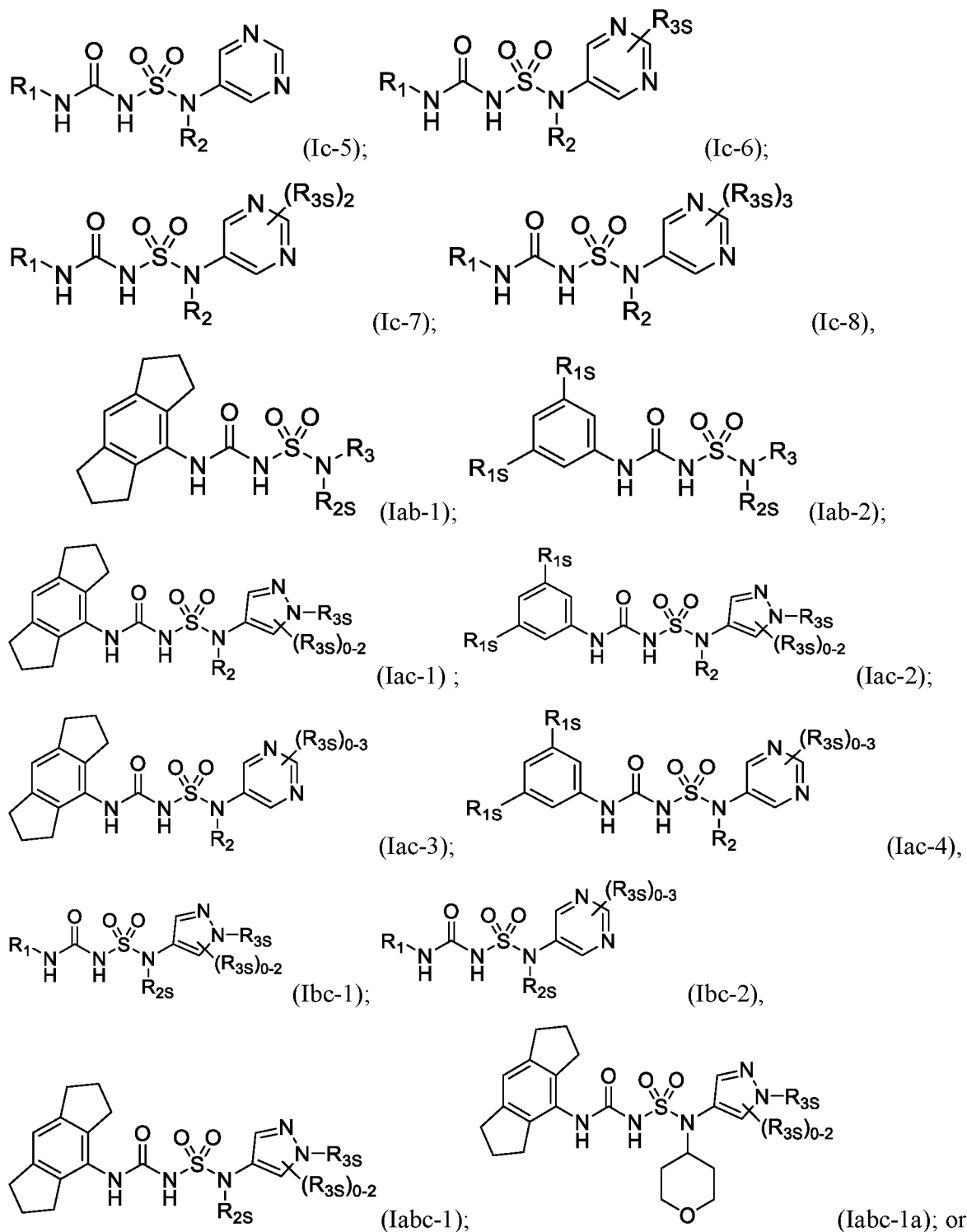


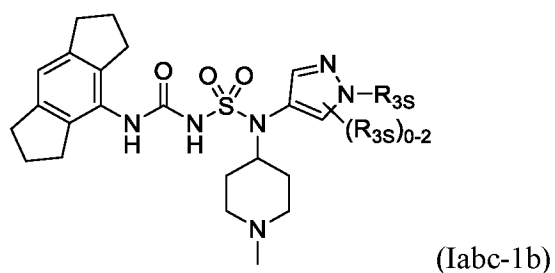
6. The compound of any one of claims 1 to 5, wherein at least one **R_{3S}** is C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN;

optionally wherein at least one R_{3S} is , methyl, ethyl, isopropyl, -CH₂OCH₃, -CH₂CF₃, -CH₂CH₂OCH₃, -CH₂CN, CH₂CF₂, or -CH₂CH₂N(CH₃)₂.

7. The compound of any one of the preceding claims, wherein the compound is of Formula (Ia-1), (Ia-2), (Ib-1), (Ib-2), (Ib-3), (Ib-1a), (Ib-1b), (Ib-1c), (Ic-1), (Ic-2), (Ic-3), (Ic-4), (Ic-5), (Ic-6), (Ic-7), (Ic-8), (Iab-1), (Iab-2), (Iac-1), (Iac-2), (Iac-3), (Iac-4), (Ibc-1), (Ibc-2), (Iabc-1), (Iabc-1a), or (Iabc-1b):

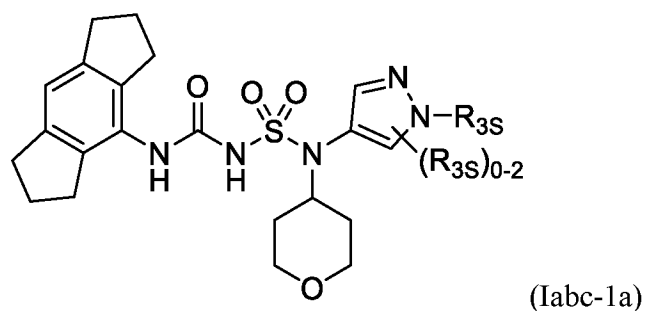






or a solvate, or pharmaceutically acceptable salt thereof, wherein **R₁**, **R_{2S}**, **R₂**, and **R₃** are as described herein.

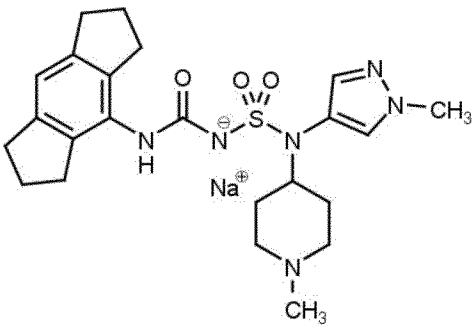
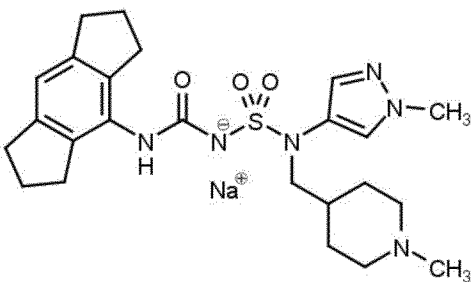
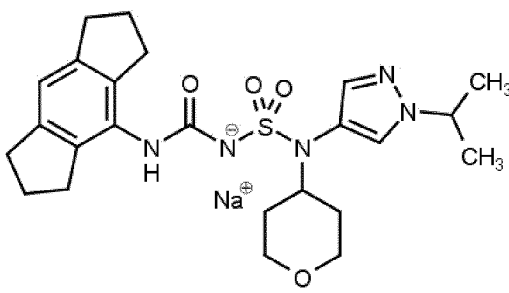
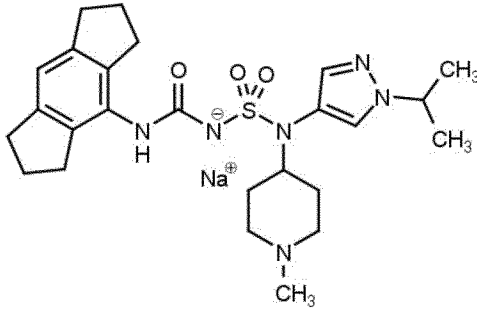
8. The compound of any one of claims 1 to 7, wherein the compound is of Formula (Iabc-1a):



or a solvate, or pharmaceutically acceptable salt thereof, wherein **R_{3S}** is as described herein.

9. The compound of any one of the claims 1 to 8, being selected from:

Compound No.	Structure
1	

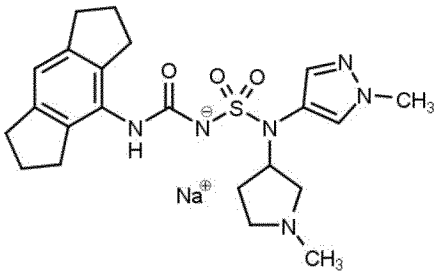
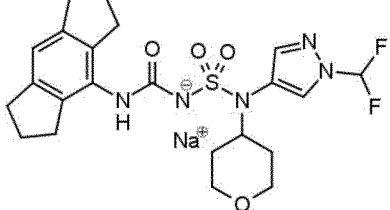
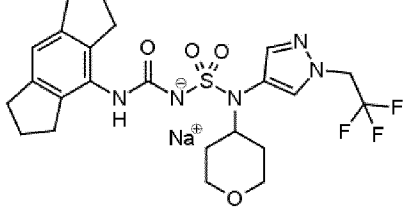
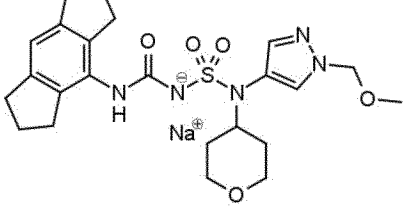
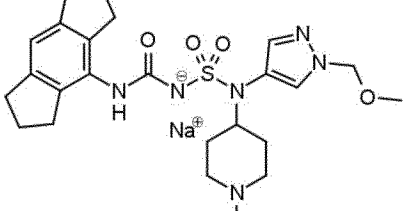
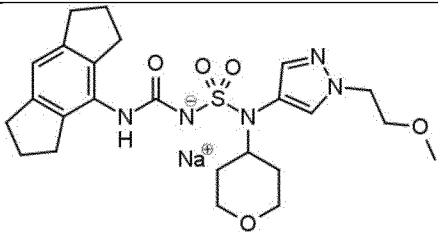
Compound No.	Structure
2	
3	
4	
5	

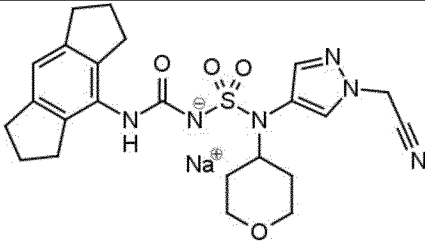
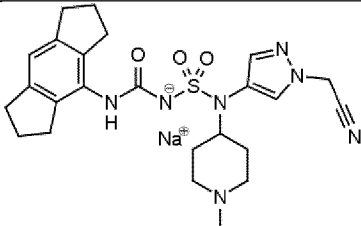
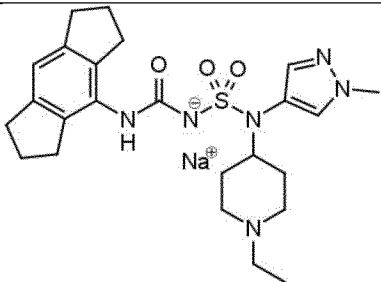
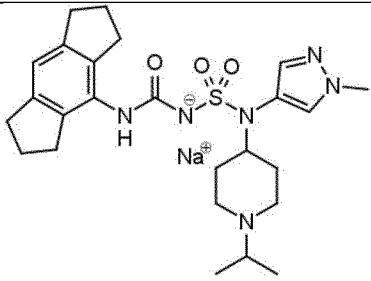
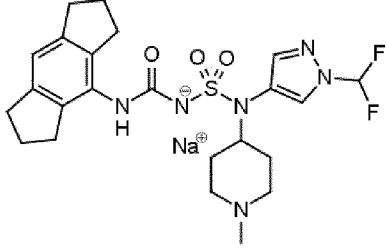
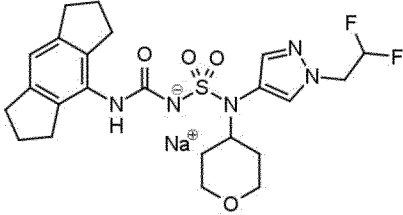
Compound No.	Structure
6	 <chem>CC1=CN(C1)N(C2CCOCC2)S(=O)(=O)N(C(=O)N3C4C=CC5C4C=CC5C3)C6=CC=CC=C6</chem>
7	 <chem>CC1CCN(C1)N(C2CC=CN2)S(=O)(=O)N(C(=O)N3C4C=CC5C4C=CC5C3)C6=CC=CC=C6</chem>
8	 <chem>CC1=CN(C1)N(C2CCOC2)S(=O)(=O)N(C(=O)N3C4C=CC5C4C=CC5C3)C6=CC=CC=C6</chem>
9	 <chem>CC1=CN(C1)N(C2CCOCC2)S(=O)(=O)N(C(=O)N3C4C=CC5C4C=CC5C3)C6=CC=CC=C6</chem>
10	 <chem>CC1=CN(C1)N(C2CCOCC2)S(=O)(=O)N(C(=O)N3C4C=CC5C4C=CC5C3)C6=CC=CC=C6</chem>

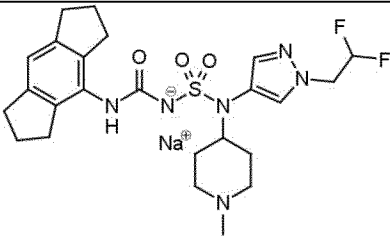
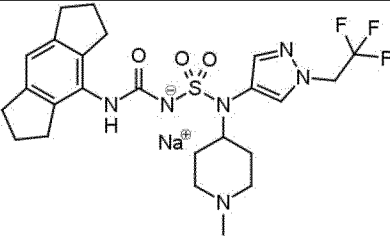
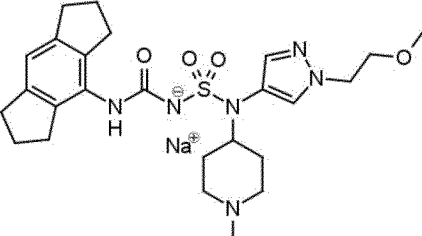
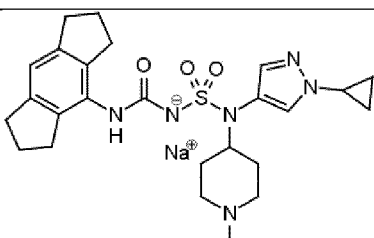
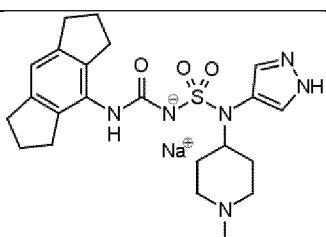
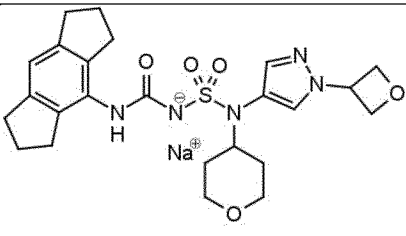
Compound No.	Structure
11	 <chem>Cc1nc(C)c(NC(=O)Nc2c3c(c4ccccc4c2)C5CCCC5)cc1S(=O)(=O)N([Na+])c6ccc(OC)cc6</chem>
12	 <chem>Cc1cc2nc(O)nn2c1N(S(=O)(=O)N([Na+])c3ccc(OC)cc3)C(=O)Nc4c5c(c6ccccc6c4)C7CCCC7</chem>
13	 <chem>c1cc2nc(O)nn2c1N(S(=O)(=O)N([Na+])c3ccc(OC)cc3)C(=O)Nc4c5c(c6ccccc6c4)C7CCCC7</chem>
14	 <chem>Cc1nc(C)c(NC(=O)Nc2c3c(c4ccccc4c2)C5CCCC5)cc1S(=O)(=O)N([Na+])c6ccncc6C</chem>

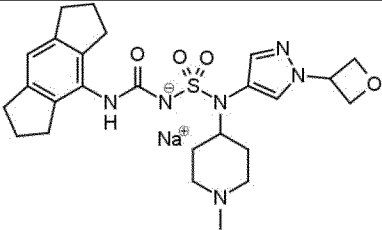
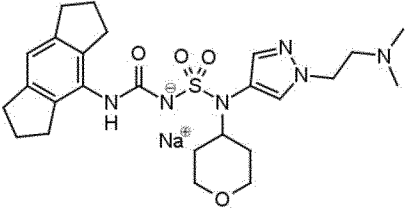
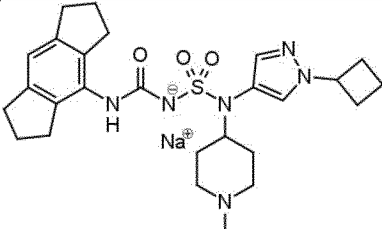
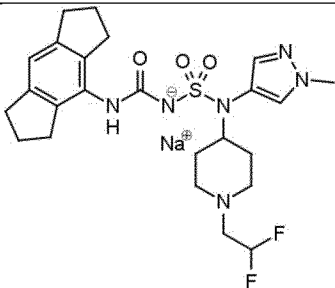
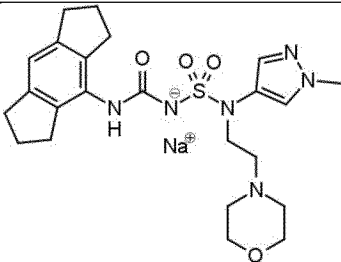
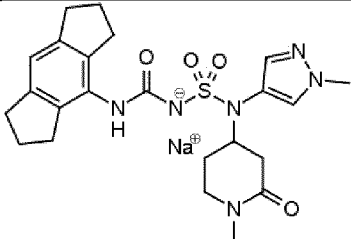
Compound No.	Structure
15	 <chem>CN1CCN(C)CC1N(S(=O)(=O)NC(=O)Nc2c3c(c4c2C5CCCC5)C6=CC=CC=C6C7CCCC7)C8=CN(C)C=N8.[Na+]</chem>
16	 <chem>CN1C=NC=C1N(S(=O)(=O)NC(=O)Nc2c3c(c4c2C5CCCC5)C6=CC=CC=C6C7CCCC7)C8CC9OC9C8.[Na+]</chem>
17	 <chem>C1=NC2=C(N1)N=CN=C2N(S(=O)(=O)NC(=O)Nc3c4c(c5c3C6CCCC6)C7=CC=CC=C7C8CCCC8O)C9CCOCC9.[Na+]</chem>
18	 <chem>Cc1cc2nnc1oc2N(S(=O)(=O)NC(=O)Nc3c4c(c5c3C6CCCC6)C7=CC=CC=C7C8CCCC8O)C9CCOCC9.[Na+]</chem>
19	 <chem>CN1C=NN=C1N(S(=O)(=O)NC(=O)Nc2c3c(c4c2C5CCCC5)C6=CC=CC=C6C7CCCC7)C8CCOCC8.[Na+]</chem>

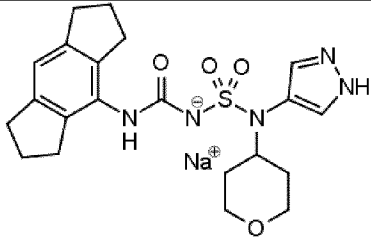
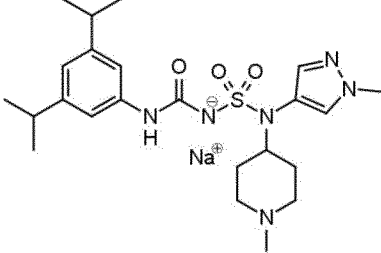
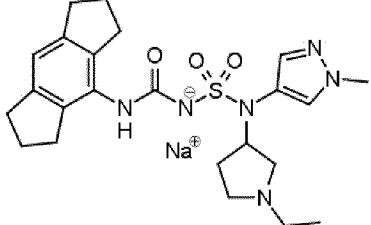
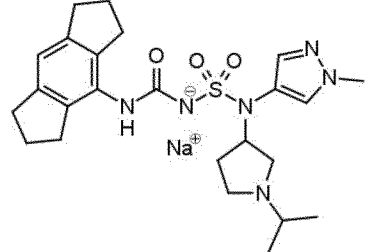
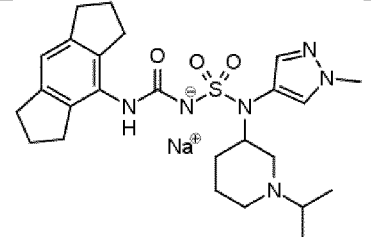
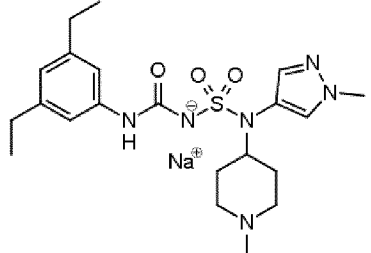
Compound No.	Structure
20	 <chem>CN1CCN(C1)S(=O)(=O)NC(=O)Nc2c3c(c4ccccc4c23)C5CCCC5</chem>
21	 <chem>CN1CCOC(C1)S(=O)(=O)NC(=O)Nc2c3c(c4ccccc4c23)C5CCCC5</chem>
22	 <chem>CN1CCOC(C1)S(=O)(=O)NC(=O)Nc2c3c(c4ccccc4c23)C5CCCC5</chem>
23	 <chem>CN1CCN(C1)S(=O)(=O)NC(=O)Nc2c3c(c4ccccc4c23)C5CCCC5</chem>

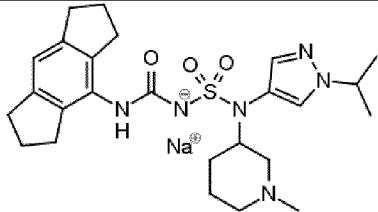
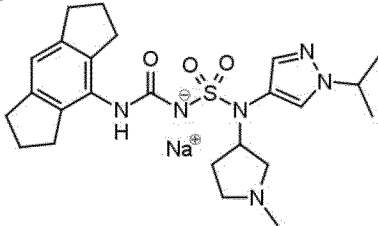
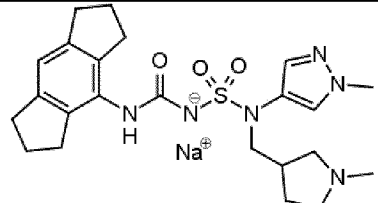
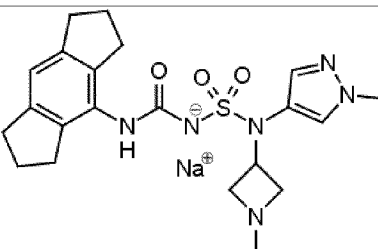
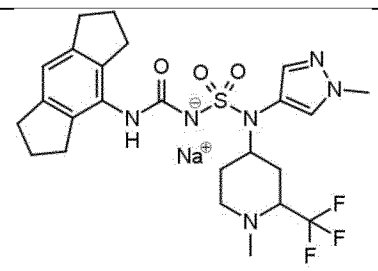
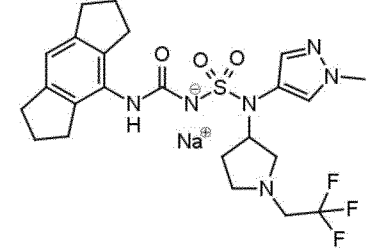
Compound No.	Structure
24	
25	
26	
27	
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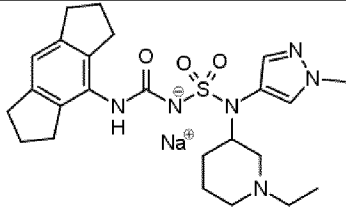
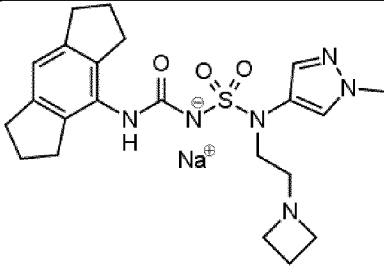
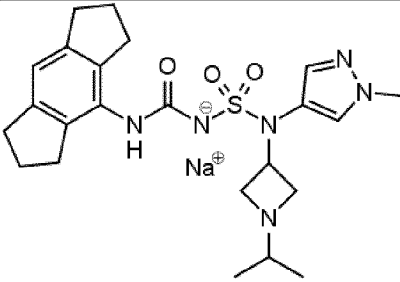
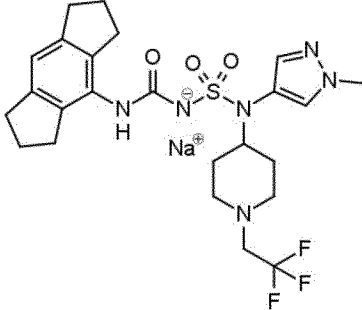
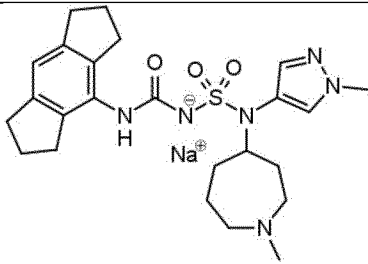
Compound No.	Structure
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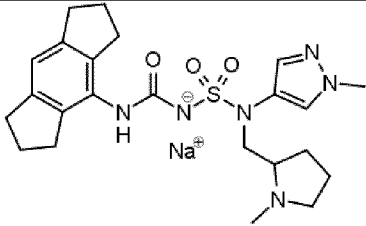
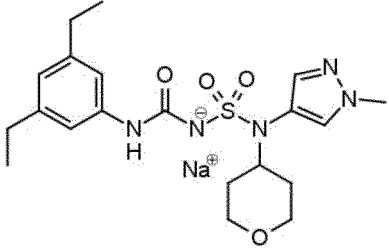
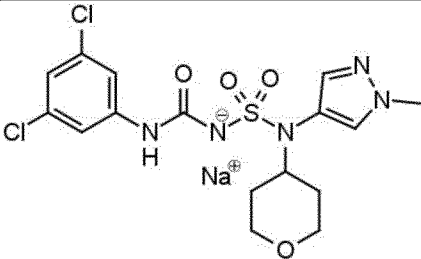
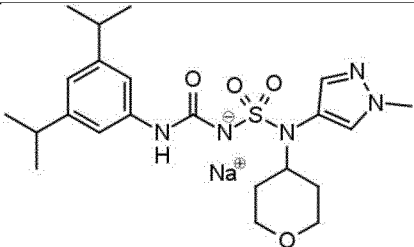
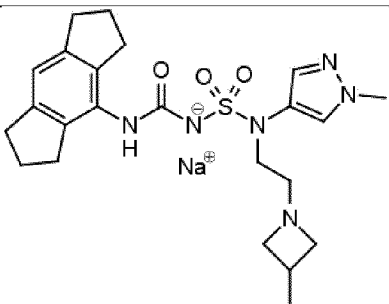
Compound No.	Structure
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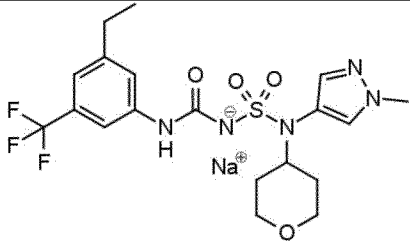
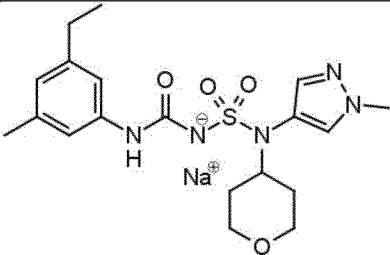
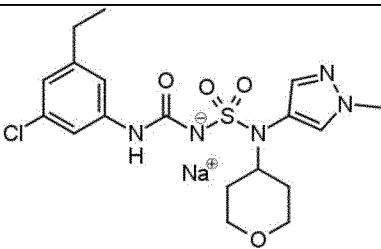
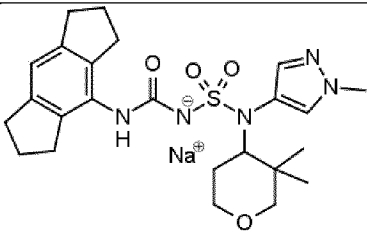
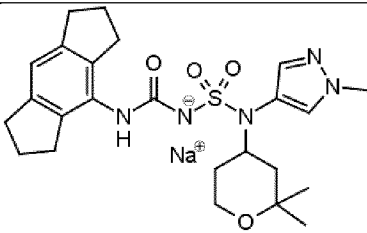
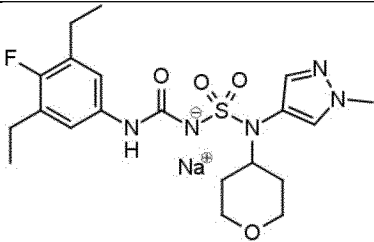
Compound No.	Structure
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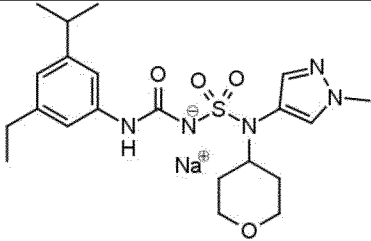
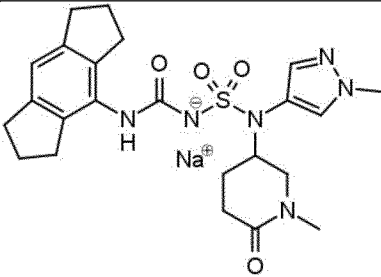
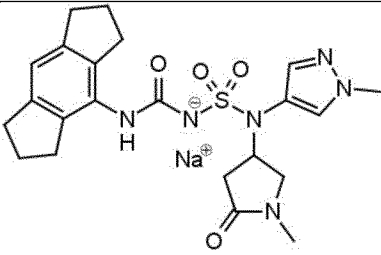
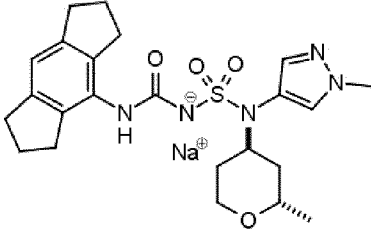
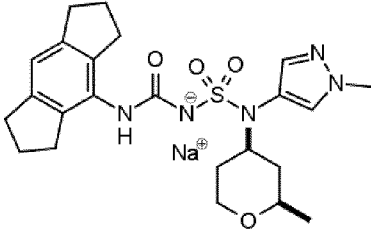
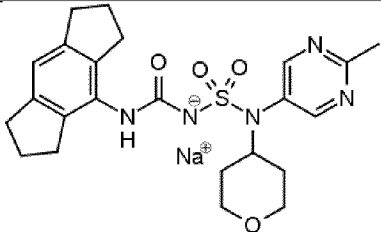
Compound No.	Structure
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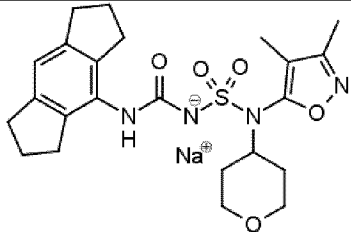
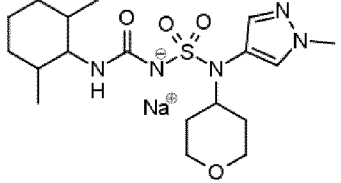
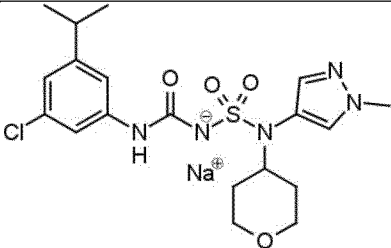
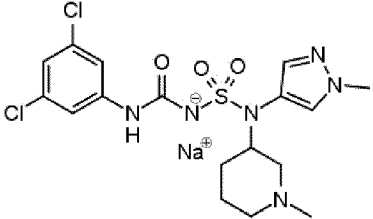
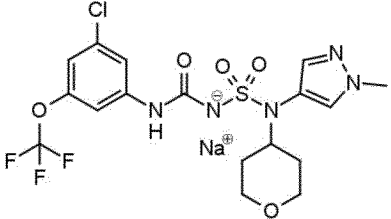
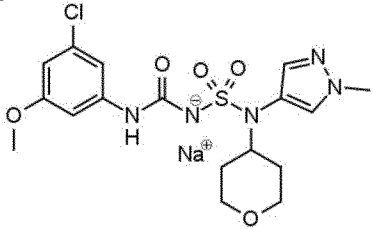
Compound No.	Structure
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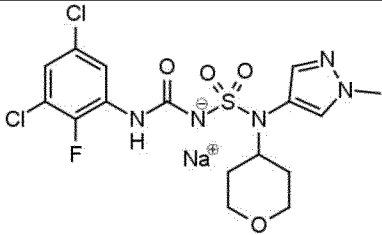
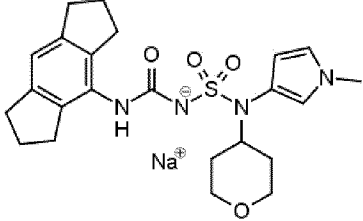
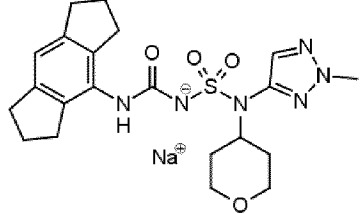
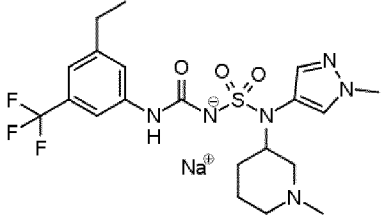
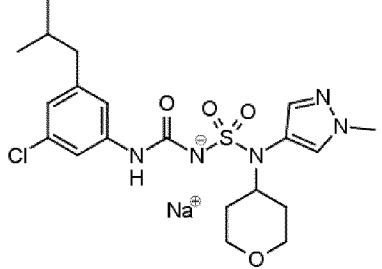
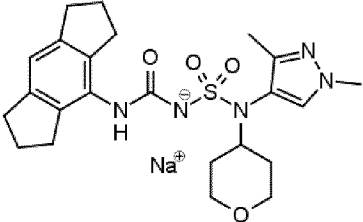
Compound No.	Structure
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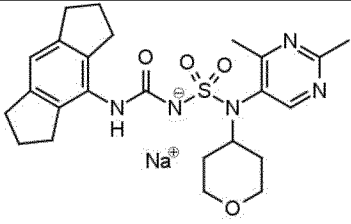
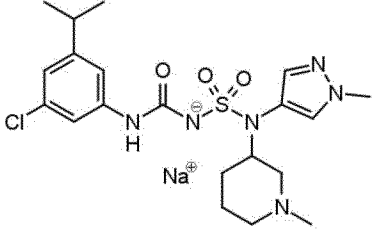
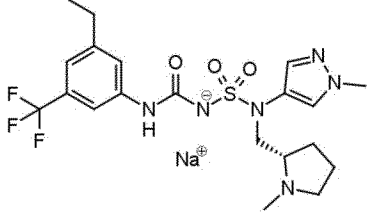
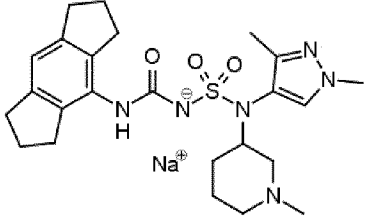
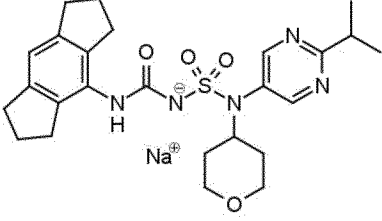
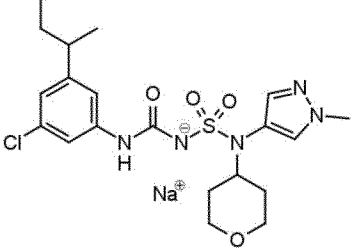
Compound No.	Structure
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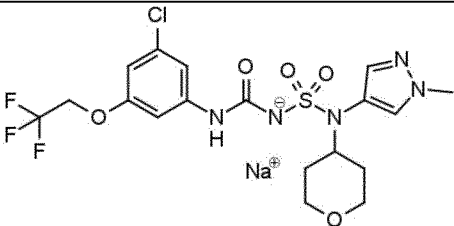
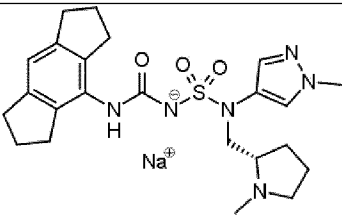
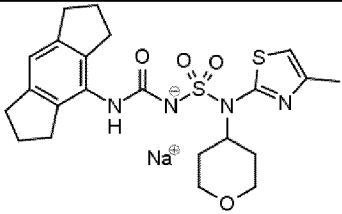
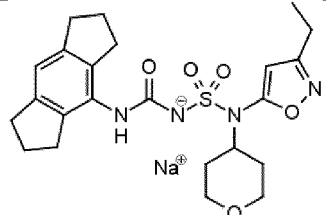
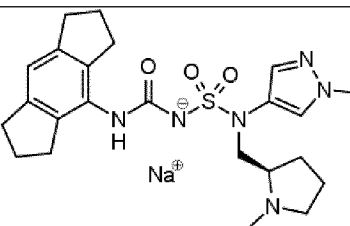
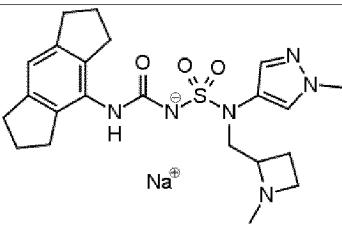
Compound No.	Structure
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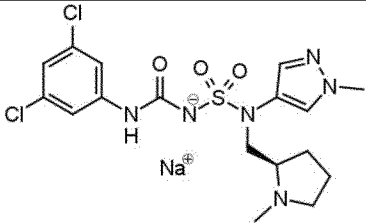
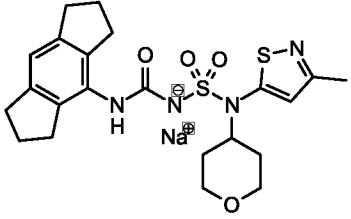
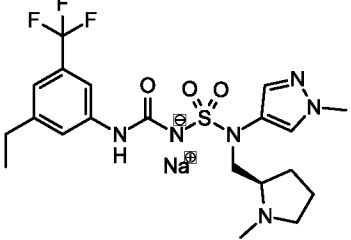
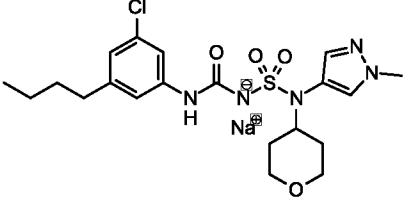
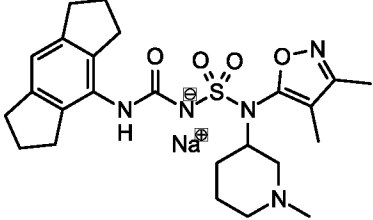
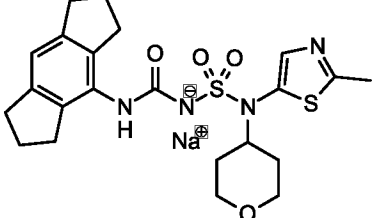
Compound No.	Structure
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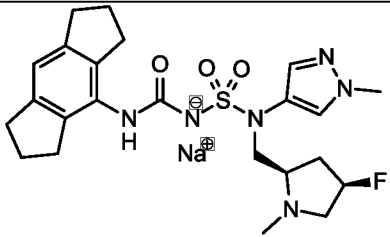
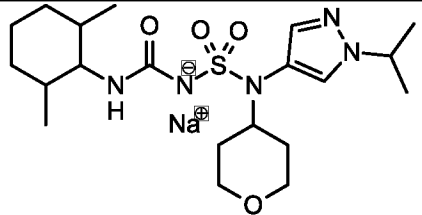
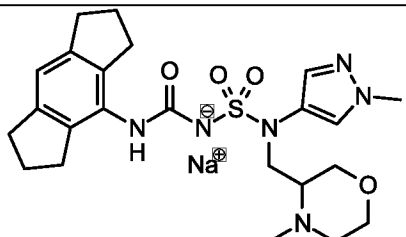
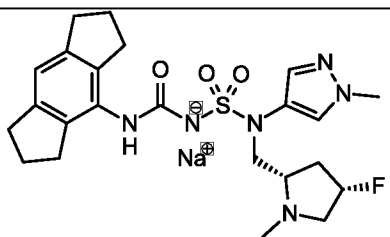
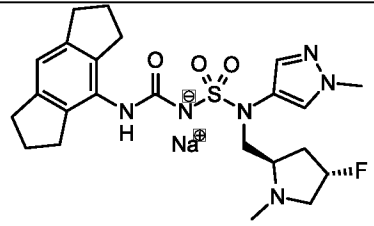
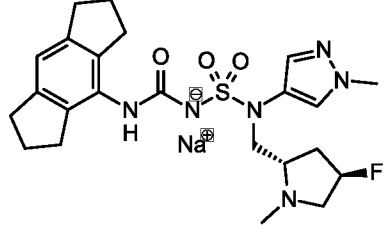
Compound No.	Structure
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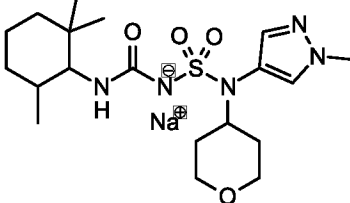
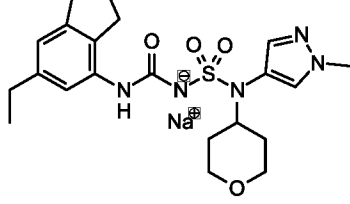
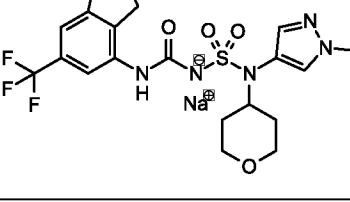
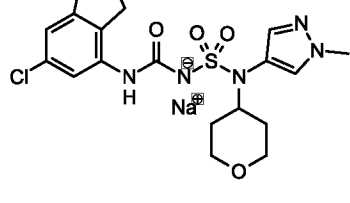
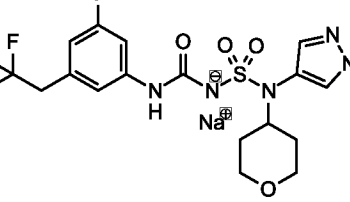
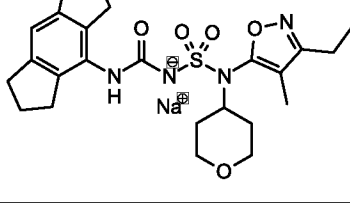
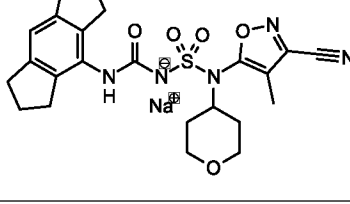
Compound No.	Structure
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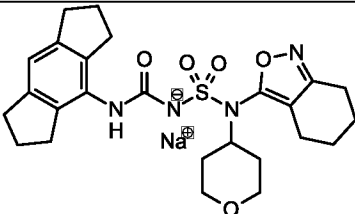
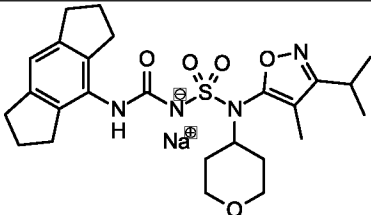
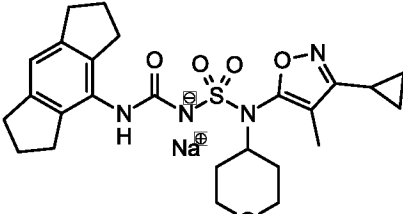
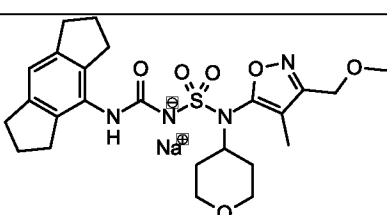
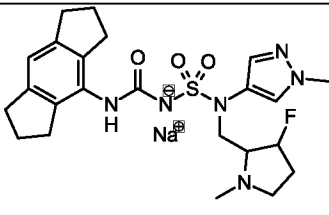
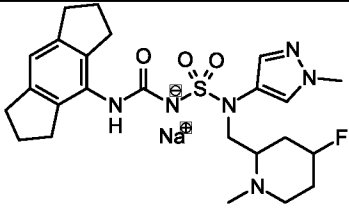
Compound No.	Structure
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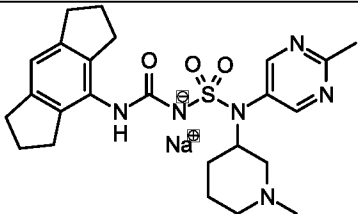
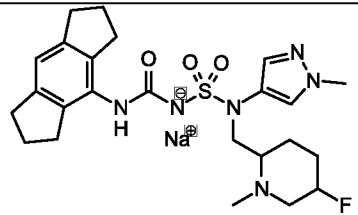
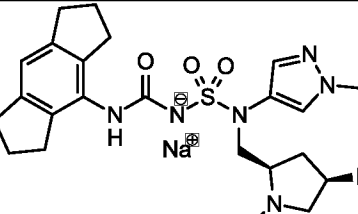
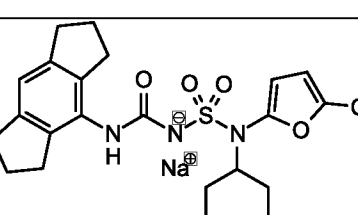
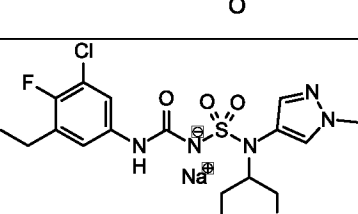
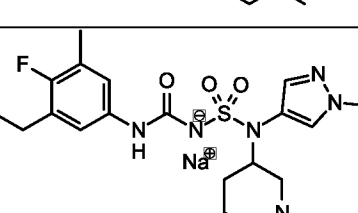
Compound No.	Structure
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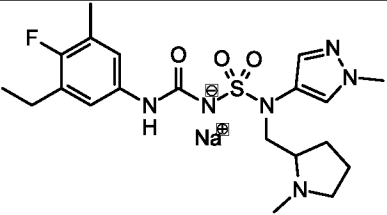
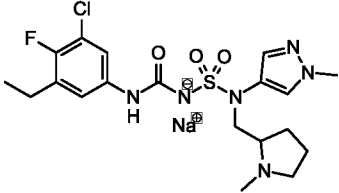
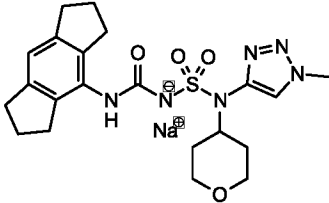
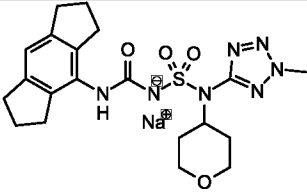
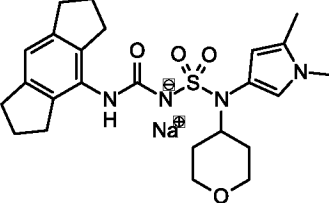
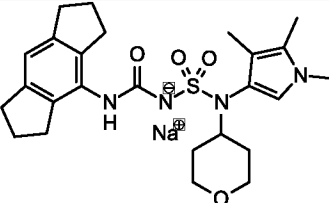
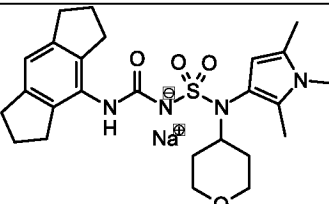
Compound No.	Structure
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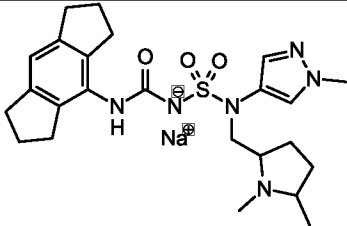
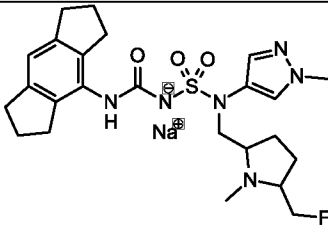
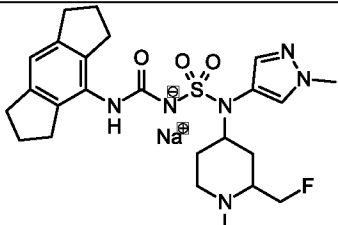
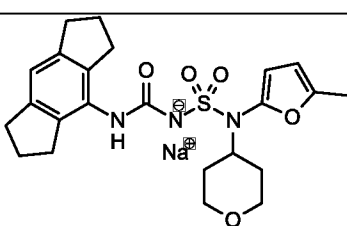
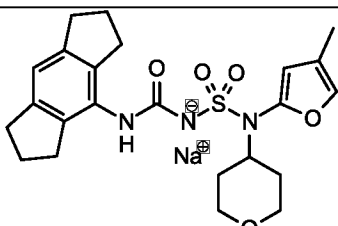
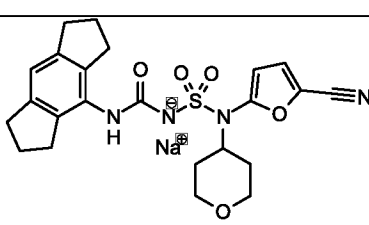
Compound No.	Structure
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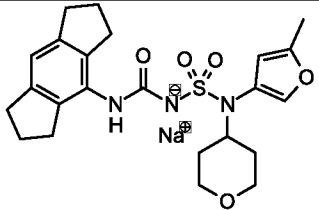
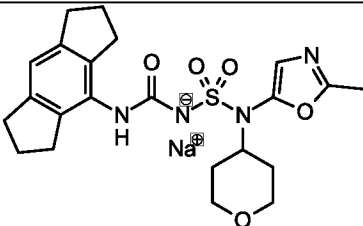
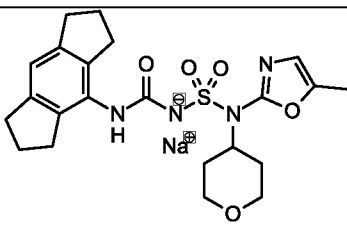
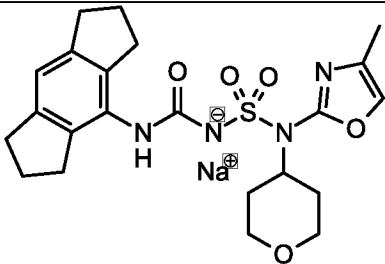
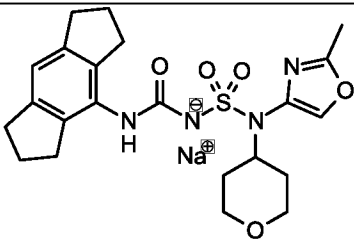
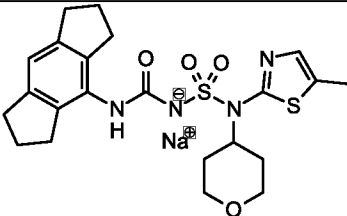
Compound No.	Structure
124	
125	
126	
127	
128	
129	
130	

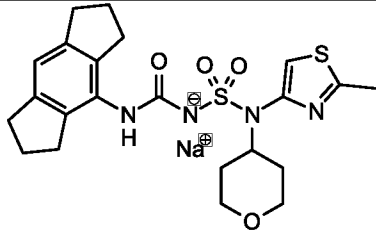
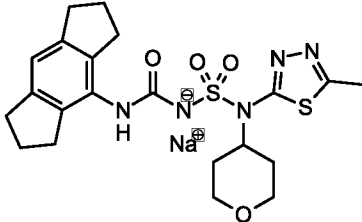
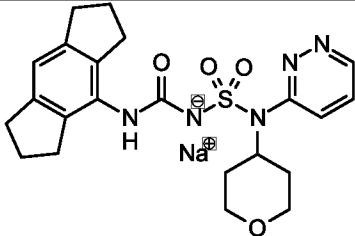
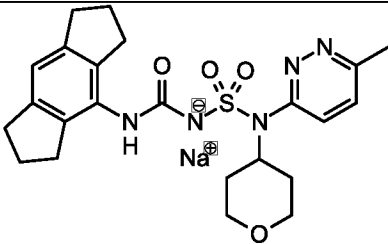
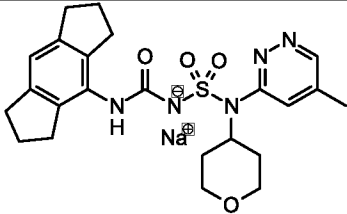
Compound No.	Structure
131	
132	
133	
134	
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136	

Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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and pharmaceutically acceptable salts thereof.

10. A pharmaceutical composition comprising the compound of any one of claims 1 to 9 or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable diluent or carrier.

11. The compound of any one of claims 1 to 9, or the pharmaceutical composition of claim 10,

for use in inhibiting inflammasome activity; optionally, the inflammasome is NLRP3 inflammasome, and the activity is *in vitro* or *in vivo*.

12. The compound of any one of claims 1 to 9, or the pharmaceutical composition of claim 10, for use in treating or preventing a disease or disorder; preferably wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder, an autoimmune disorder, a neurodegenerative disease, or cancer.

13. The compound or pharmaceutical composition for use of claim 11 or 12, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder or an autoimmune disorder; optionally, the disease or disorder is selected from cryopyrin-associated autoinflammatory syndrome (CAPS; e.g., familial cold autoinflammatory syndrome (FCAS), Muckle-Wells syndrome (MWS), chronic infantile neurological cutaneous and articular (CINCA) syndrome/ neonatal-onset multisystem inflammatory disease (NOMID)), familial Mediterranean fever (FMF), nonalcoholic fatty liver disease (NAFLD), non-alcoholic steatohepatitis (NASH), gout, rheumatoid arthritis, osteoarthritis, Crohn's disease, chronic obstructive pulmonary disease (COPD), chronic kidney disease (CKD), fibrosis, obesity, type 2 diabetes, multiple sclerosis, dermatological diseases (e.g., acne) and neuroinflammation occurring in protein misfolding diseases (e.g., Prion diseases).

14. The compound or pharmaceutical composition for use of claims 11 or 12, wherein disease or disorder is a neurodegenerative disease; optionally, the disease or disorder is Parkinson's disease or Alzheimer's disease.

15. The compound or pharmaceutical composition for use of claims 11 or 12, wherein the disease or disorder is cancer; optionally, the cancer is metastasising cancer, gastrointestinal cancer, brain cancer, skin cancer, non-small-cell lung carcinoma, head and neck squamous cell carcinoma or colorectal adenocarcinoma.